SIH 2026 Proposed Solution & Architecture Report

AEGIS Sovereign Intelligence Report | Generated: 2026-09-02 15:21:07 UTC | Classification: **CONFIDENTIAL INTERNAL**

EXECUTIVE SUMMARY

Aegis proposes a sovereign on-premise AI workbench, ‘Sovereign AI Workbench,’ leveraging open-weight multimodal LLMs for confidential industrial work. The solution emphasizes 100% on-premise processing, intelligent model routing, agentic workflows (Plan → Execute → Verify → Re-plan), multimodal understanding (PDF, Image, Drawing), local RAG, and sandboxed tool execution. The core innovation lies in dynamic model loading and a modular architecture designed for secure, traceable, and adaptable AI within restricted environments.

KEY FINDINGS

The ‘Sovereign AI Workbench’ addresses the need for secure AI processing within industrial environments. It utilizes a decentralized architecture with local execution, eliminating cloud AI API calls and ensuring data remains within the organization’s network. Key features include an Agentic AI system that automates tasks through a Plan-Execute-Verify-Re-plan loop, multimodal document processing (text, PDFs, scans, images, drawings), a Model Registry for dynamic model management, and a secure web UI with authentication and audit logging. The system employs quantization to optimize model performance and utilizes a dynamic model loading mechanism to select the appropriate model for each task, increasing efficiency. The system also uses OCR and Vision models.

DETAILED ANALYSIS

The proposed solution incorporates several technological components. The core agent controller processes queries, planning tasks, and selecting models. The document workflow handles PDFs, scans, and text summarization via RAG. The coding workflow supports code generation, execution, and debugging within a sandboxed environment. The multimodal workflow processes images and drawings utilizing OCR and vision models. A Model Registry manages models, employing dynamic load/unload capabilities, and prioritizing task-aware model selection via quantization. The system's modular design allows for easy addition or modification of AI models without affecting the core application. The system is designed for deployment across workstations and enterprise environments, with a focus on air-gapped and confidential data protection. The system relies on GPU management to handle resource contention.

RISKS AND OPERATIONAL ISSUES

Several potential risks and operational challenges were identified. GPU limits and memory contention could impact performance. The OCR + VLM + Document pipeline requires careful optimization. Local processing with network isolation introduces potential for network latency and requires robust security measures. RAG + Source Grounding + Validation needs to prevent hallucinations and ensure data integrity. Complex document processing and the potential for model hallucination represent key concerns. Data leakage prevention is critical due to the sensitive nature of the data processed. The system depends on a robust audit trail and user role management to maintain control and traceability.

RECOMMENDATIONS

To mitigate risks and ensure successful implementation, Aegis recommends prioritizing GPU performance optimization, rigorous testing of the OCR + VLM pipeline, robust network security protocols, and the development of validation strategies to combat hallucination. Continued monitoring of resource utilization is essential. Investment in user training focused on agentic workflows and data governance is also advised. The team should continue research into existing sovereign AI solutions (Siemens, Palantir, IBM, Aleph Alpha) to refine their approach and stay abreast of industry best practices.

AUTHORITATIVE SOURCES & CITATIONS

Document Name	Referenced Pages	Relevance Status
SIH2026ppt.pdf	Page(s) 1, 2, 3, 4, 5, 6, 7	High

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